

Credit Card Fraud Detection Using Machine Learning

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1. Problem Description

The exponential growth of digital transactions, online banking, and e-commerce platforms has significantly increased the risk of credit card fraud. Fraudulent activities result in billions of dollars in losses annually and pose serious threats to financial institutions and customers.

A major challenge in fraud detection is the extreme class imbalance present in datasets. In real-world datasets, fraudulent transactions account for less than 0.2% of total transactions. This makes traditional machine learning models ineffective, as they tend to bias toward the majority class.

The objective of this project is to develop a robust machine learning-based fraud detection system that:

Accurately classifies transactions as fraudulent or legitimate

Handles imbalanced data efficiently

Minimizes false positives and false negatives

Provides reliable predictions in real-time scenarios

The system primarily uses the K-Nearest Neighbors (KNN) algorithm along with proper preprocessing and evaluation techniques.

2. Architecture

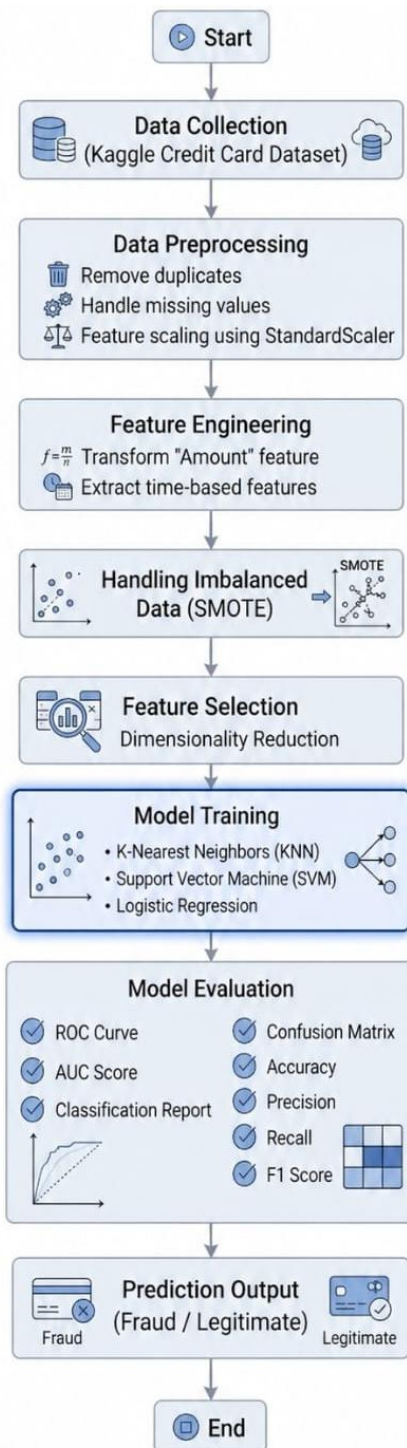


Fig.1 Sample flow diagram.

3.Models Used(Applied/ Planned) Give a brief introduction of the models used

In this project, multiple machine learning models are studied and analyzed for detecting fraudulent transactions. The following three models are considered:

1. K-Nearest Neighbors (KNN)

KNN is a supervised learning algorithm used for classification. It works based on similarity between data points.

Working:

Select value of K

Calculate distance between data points using Euclidean distance

Select K nearest neighbors

Assign class based on majority voting

Advantages:

Simple and easy to implement

No training phase (lazy learner)

Effective for small datasets

Disadvantages:

Slow for large datasets

Sensitive to noise and irrelevant features

Requires proper feature scaling

2. Logistic Regression

Logistic Regression is a statistical classification algorithm used for binary classification problems.

It predicts the probability that a given input belongs to a certain class using the sigmoid function.

Formula:

Working:

Takes weighted sum of input features

Applies sigmoid function

Outputs probability between 0 and 1

Classifies based on threshold (0.5)

Advantages:

Fast and efficient

Works well for linearly separable data

Easy to interpret

Disadvantages:

Cannot handle complex non-linear relationships

Sensitive to multicollinearity

3. Support Vector Machine (SVM)

SVM is a powerful supervised learning algorithm used for classification and regression.

It finds the optimal hyperplane that separates different classes with maximum margin.

Working:

Maps data into high-dimensional space

Finds optimal boundary (hyperplane)

Uses support vectors (critical points)

Advantages:

Works well for high-dimensional data

Effective in complex classification tasks

Robust to overfitting

Disadvantages:

Computationally expensive

Difficult to tune parameters

Less effective for very large datasets

4. Evaluation metrics used

Due to class imbalance, multiple evaluation metrics are used:

Accuracy – Overall correctness

Precision – Reduces false fraud detection

Recall – Detects actual fraud cases

F1 Score – Balance between precision & recall

Confusion Matrix – Detailed prediction analysis

Auc score- measures binary classifiers ability to distinguish between classes

Roc curve- is a graph visualization binary classification model performance across all threshold plotting the true positive rate versus false positive rate

Classification report- evaluation metric that displays precision recall f1 score and support for each class in a classification model

5. Preliminary Results and Visualizations

The following visualizations were generated:

Correlation heatmap

Class imbalance distribution graph

Feature importance visualization

Observations:

Strong correlations exist between certain features

Fraud class is significantly underrepresented

Visualization helps in understanding patterns

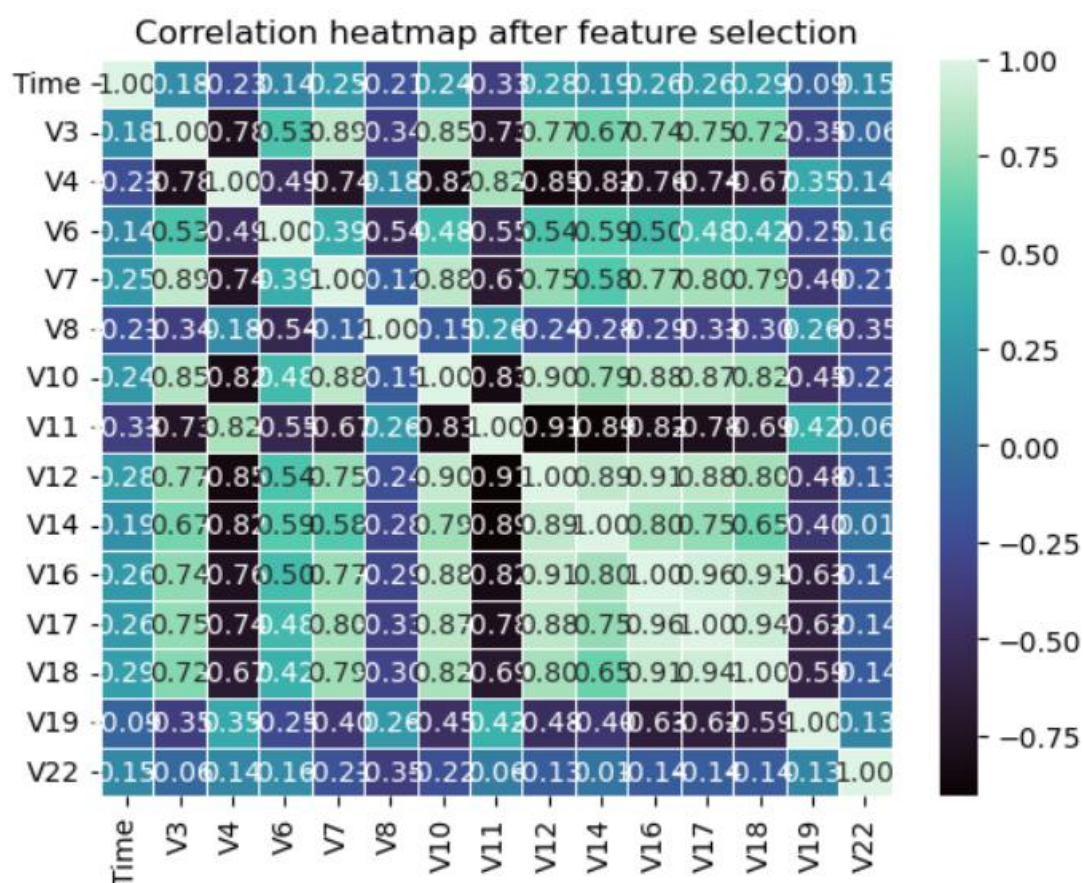


Fig.2 Heatmap showing correlation.

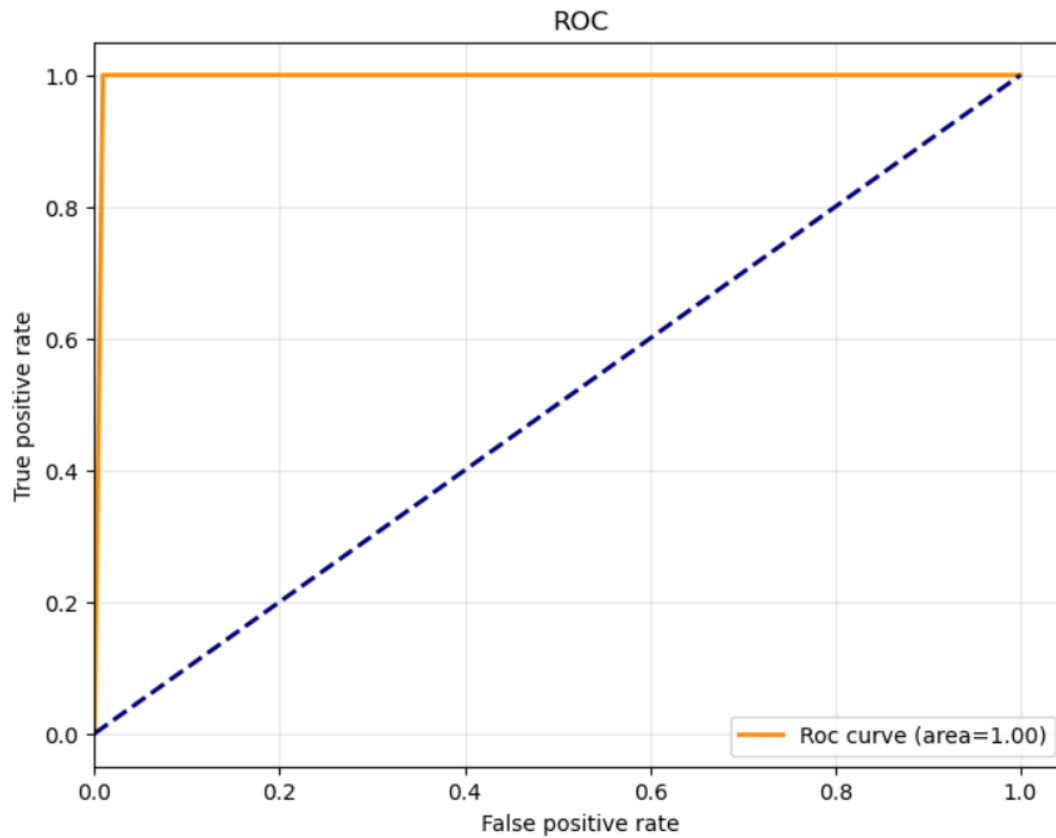


Fig3. Roc curve of knn

	precision	recall	f1-score	support
0	0.97	0.99	0.98	56596
1	0.99	0.96	0.98	55446
accuracy			0.98	112042
macro avg	0.98	0.98	0.98	112042
weighted avg	0.98	0.98	0.98	112042

Fig4. Logistic regression

	precision	recall	f1-score	support
0	1.00	0.99	1.00	56596
1	0.99	1.00	1.00	55446
accuracy			1.00	112042
macro avg	1.00	1.00	1.00	112042
weighted avg	1.00	1.00	1.00	112042

Fig5. KNN

	precision	recall	f1-score	support
0	0.98	0.99	0.98	56596
1	0.99	0.98	0.98	55446
accuracy			0.98	112042
macro avg	0.98	0.98	0.98	112042
weighted avg	0.98	0.98	0.98	112042

Fig6. SVM

```
Model loaded successfully
['Time', 'V3', 'V4', 'V6', 'V7', 'V8', 'V10', 'V11', 'V
12', 'V14', 'V16', 'V17', 'V18', 'V19', 'V22']
Enter above mentioned features: 0 2.536346738 1.3781552
24 0.462387778 0.239598554 0.098697901 0.090794172 -0.5
51599533 -0.617800856 -0.311169354 -0.470400525 0.2079
71242 0.02579058 0.40399296 0.277837576
0:Legitimate Transaction
```

Fig 6.1

```
Model loaded successfully
['Time', 'V3', 'V4', 'V6', 'V7', 'V8', 'V10', 'V11', 'V
12', 'V14', 'V16', 'V17', 'V18', 'V19', 'V22']
Enter above mentioned features: 406 -1.609850732 3.9979
05588 -1.426545319 -2.537387306 1.391657248 -2.77227214
5 3.202033207 -2.899907388 -4.289253782 -1.14074718 -2.
830055675 -0.016822468 0.416955705 -0.035049369
1:Fraudulent Transaction
```

Fig 6.2

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- [Kaggle Dataset](#)
- [Scikit-learn Documentation](#)
- [GeeksforGeeks](#)